Yunzhe Daniel **Wang**

LLM Agents · Multi-Agent Systems · World Models

Los Angeles, CA

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Education

University of Southern California — *Los Angeles, CA*

PhD in Computer Science · Advised by Volkan Ustun and Gale M. Lucas — *Aug 2024 – Expected 2028*

- Research on **self-improving LLM agents**, **LLM-driven social simulation**, and **multi-agent video world models**

Columbia University — *New York, NY*

MS in Computer Science, Machine Learning Track · Advised by Hod Lipson — *Aug 2021 – May 2023*

University of Southern California — *Los Angeles, CA*

BS in Computational Neuroscience, BA in Applied Mathematics, Minor in Computer Science — *Aug 2017 – May 2021*

Experience

Amazon Web Services (AWS) – Amazon Quick, Agentic AI Org — *Santa Clara, CA*

Applied Scientist Intern — *Jun 2026 – Present*

- ***Mixed User–Agent Co-Evolution***
  - Proposed and led research on a scalable **multi-granularity memory** through which a population of users and heterogeneous LLM **coding agents** learn from one another's experience. Submitted to NAACL 2027.
  - Designed the memory to decide which users and agents each learned lesson should reach, with **privacy-first** sharing that keeps private information and personal preferences within their owner's scope.
  - Built a **harness-agnostic evaluation platform** for coding agents such as Claude Code and Codex that measures utility, privacy, and personalization jointly as users and agents co-evolve.
- ***Agentic RL Environment Synthesis***
  - Leading research on **cross-environment, long-horizon tasks** that chain several synthesized, stateful services over many steps, giving agentic RL a harder, more realistic curriculum.
  - Led the core system, which synthesizes complex, executable tool-use environments from service documentation for **LLM agent post-training** with SFT and RLVR.

University of Southern California — *Los Angeles, CA*

Graduate Research Assistant — *Aug 2024 – Present*

- Lead research on **LLM agents, social simulation, multi-agent video world models, and agentic RL**. First-author papers at ACL 2026, EMNLP 2025, and WSC 2025. Mentored 4 students to papers at NeurIPS 2026 Creative AI, ACL 2026, and an IJCAI-ECAI 2026 workshop.
- ***LLM Agents and Social Simulation***
  - Developed *PersonaEvolve* (EMNLP 2025), a gradient-free **LLM-as-optimizer** that rewrites the personas of generative agents to match expert crowd behavior, cutting divergence from expert data by 84% and beating explicit instruction on average without losing contextual realism.
  - Built the Unity simulator behind it and a **browser-native successor** in which experts review 100+ LLM agents acting in real campus buildings, through memory-grounded agent interviews and a validated LLM behavior classifier.
  - Developed *IPA-Bench*, revealing that much of LLMs' apparent skill at inferring a **specific real person's perspective** is not person-specific.
  - Contributed to open **agent benchmarks** *Agent's Last Exam* (UC Berkeley), *Science Trajectory* (UCSD), and *Operation Research Bench* (Cornell).
- ***Multi-Agent Video World Models***
  - Originated a research direction on **large-scale multi-agent video world models**. Led and conceived *FlockWorld* (NeurIPS 2026 Creative AI), mentored its undergraduate first author, and designed its **diffusion transformer** that jointly generates 10 agents' first-person views.
  - Led a large-scale mechanism study showing that **diffusion forcing** cuts cross-view error 4.4× in long autoregressive rollouts, beating the full MIRA multi-agent recipe. Designed ground-truth-free cross-view metrics and now designing a larger successor model.
- ***Multi-Agent Video Understanding***
  - Created *GameplayQA* (ACL 2026), first agent-centric, multi-POV-synchronized video benchmark, revealing where today's multimodal LLMs fall short of humans, with errors growing as synchronized views are added.
  - Proposed *CECL* (X-Ego), a **cross-egocentric contrastive** objective that aligns teammates' first-person views for team-level awareness, and built the *X-Ego-CS* dataset of all 10 players in expert Counter-Strike 2 matches. A follow-up roughly tripled teammate and enemy localization.
- ***Agentic, Multi-Objective, and Multi-Agent RL***
  - Developed *Steering Room*, making steering in **preference-conditioned agentic RL** measurable with **exact regret** on a prompt-injection mailbox. A single small RL-trained policy spans the **Pareto front** of utility, security, and efficiency, re-weighted at inference without retraining, and outperforms fixed-behavior baselines.
  - Co-developed *GraphAllocBench* (MODeM @ IJCAI-ECAI 2026), a graph-based benchmark for preference-conditioned multi-objective RL whose new reliability and preference-alignment metrics expose failures that hypervolume misses.

– Developed *DECOY* (WSC 2025), a data-driven **multi-agent RL** simulator of CS:GO with combat models learned from professional matches. Replaying human rounds reproduced 91% of match outcomes, making it a realistic testbed for strategic multi-agent planning.

Bubble.io
New York, NY

AI Software Engineer II — *Jul 2023 – Jul 2024*

- **Founding engineer** of the Bubble AI team. Took Bubble's first generative AI product, AI Page Designer, from zero to public beta launch (Jun 2024), generating responsive, editable no-code web pages from text.
- Designed the core **LLM multi-agent architecture** and a domain-specific language through which agents plan, communicate, and build full-stack web apps with users. It became the backend of the Bubble AI service.

Columbia University – Creative Machines Lab
New York, NY

Research Assistant — *Sep 2021 – May 2023*

- Built deep encoder–decoder models that let a reconfigurable legged robot infer its own body configuration from motion alone and generalize to unseen morphologies (IROS 2024, Oral).
- Built the **inverse self-model** and human-to-robot retargeting for *Emo*, a robot face that anticipates and co-expresses human smiles in real time (Science Robotics 2024), later reused for learned lip sync (Science Robotics 2026, cover).

USC Institute for Creative Technologies
Los Angeles, CA

Student Researcher — *Feb 2020 – Aug 2021*

- Built RL-trained graph-transformer **route optimization** for an AI teammate in DARPA ASIST search-and-rescue missions (first author, AAAI FSS, Springer 2023), later used to decide when the agent should step in to help a human team (ICAART 2024).
- Contributed graphical-model components to the (Py)Sigma cognitive architecture.

Chinese Academy of Sciences – Institute of Computing Technology
Beijing, China

Research Engineer Intern — *May 2019 – Aug 2019*

- Built a rule-based **NLP** system with heuristic and data-mined rules that improved Chinese part-of-speech tagging and knowledge extraction.

Publications

Yunzhe Wang is underlined. † marks a student mentored by Yunzhe.

1. Yunzhe Wang, Runhui Xu†, Kexin Zheng, Tianyi Zhang, Jayavibhav Kogundi, Soham Hans, Volkan Ustun. "**GameplayQA: A Benchmarking Framework for Decision-Dense POV-Synced Multi-Video Understanding of 3D Virtual Agents**." *Annual Meeting of the Association for Computational Linguistics (ACL 2026)*. Link
2. Meryem Koksal†, Yunzhe Wang, Volkan Ustun. "**FlockWorld: Toward Population-Scale Multi-Agent Video World Models through Artificial Life Simulation**." *Conference on Neural Information Processing Systems (NeurIPS 2026), Creative AI Track*. Link
3. Zhiheng Jiang†, Yunzhe Wang, Ryan Marr†, Ellen Novoseller, Benjamin T. Files, Volkan Ustun. "**GraphAllocBench: A Flexible Benchmark for Preference-Conditioned Multi-Objective Policy Learning**." *MODeM Workshop at IJCAI-ECAI 2026*. Link
4. Yuhang Hu, Jiong Lin, Judah Goldfeder, Philippe Wyder, Yifeng Cao, Steven Tian, Yunzhe Wang, Jingran Wang, Mengmeng Wang, Jie Zeng, Cameron Mehlman, Yingke Wang, Delin Zeng, Boyuan Chen, Hod Lipson. "**Learning Realistic Lip Motions for Humanoid Face Robots**." *Science Robotics 11(110), 2026, Cover Article*. Link
5. Yunzhe Wang, Gale M. Lucas, Burcin Becerik-Gerber, Volkan Ustun. "**Implicit Behavioral Alignment of Language Agents in High-Stakes Crowd Simulations**." *Conference on Empirical Methods in Natural Language Processing (EMNLP 2025), Main*. Link
6. Yunzhe Wang, Volkan Ustun, Chris McGroarty. "**A Data-Driven Discretized CS:GO Simulation Environment to Facilitate Strategic Multi-Agent Planning Research**." *Winter Simulation Conference (WSC 2025)*. Link
7. Volkan Ustun, Soham Hans, Rajay Kumar, Yunzhe Wang. "**Abstracting Geo-specific Terrains to Scale Up Reinforcement Learning**." *Interservice/Industry Training, Simulation, and Education Conference (I/ITSEC 2024)*. Link
8. Yuhang Hu, Yunzhe Wang, Ruibo Liu, Zhou Shen, Hod Lipson. "**Reconfigurable Robot Identification from Motion Data**." *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2024), Oral*. Link
9. Yuhang Hu, Boyuan Chen, Jiong Lin, Yunzhe Wang, Yingke Wang, Cameron Mehlman, Hod Lipson. "**Human-Robot Facial Co-expression**." *Science Robotics 9(88), 2024*. Link

10. Volkan Ustun, Ronit Jorvekar, Nikolos Gurney, David Pynadath, Yunzhe Wang. "**Assessing Routing Decisions of Search and Rescue Teams in Service of an Artificial Social Intelligence Agent**." *International Conference on Agents and Artificial Intelligence (ICAART 2024)*. Link
11. Yunzhe Wang, Nikolos Gurney, Jincheng Zhou, David Pynadath, Volkan Ustun. "**Route Optimization in Service of a Search and Rescue Artificial Social Intelligence Agent**." *AAAI Fall Symposium Series (AAAI FSS), Springer, 2023*. Link

Preprints & Under Review

1. Yunzhe Wang, Erhan Unal, Sheng Zhang, Shabnam Ghadar, Peng Tang. "**Who Remembers What? Multi-Granularity Memory Extraction for Multi-User, Multi-Agent Interaction**." *Submitted to NAACL 2027*.
2. Runhui Xu†, Yunzhe Wang, Volkan Ustun. "**ACG: Ambiguity-Aware Conditional Generation of Team Tactics**." *Submitted to AAMAS 2027*.
3. Tianyi Zhang, Yunzhe Wang, Xiaolin Zhou, Erik Cambria, David Traum, Rui Mao. "**Toward Individualized Cognitive Simulation: Benchmarking and Diagnosing Individual Perspective Alignment**." *Submitted to AAAI 2027*.
4. Yunzhe Wang, Soham Hans, Volkan Ustun. "**X-Ego: Acquiring Team-Level Tactical Situational Awareness via Cross-Egocentric Contrastive Video Representation Learning**." *arXiv:2510.19150, 2025*. Link
5. Yunzhe Wang, Soham Hans, Volkan Ustun. "**The Allocentric Shift: Multi-Ego Contrastive Perceptual Alignment from Coordinated Behavior**." *Under review at ICLR 2027*.
6. Tianyi Zhang, Yunzhe Wang, Volkan Ustun, David Traum. "**Vision-Text Augmentation: Visual Composition as an Optimization Axis for Language Tasks**." *Manuscript in preparation, 2026*.

Mentoring & Service

Mentoring	Meryem Koksal (undergraduate, first author, NeurIPS 2026 Creative AI), Zhiheng Jiang (undergraduate, first author, IJCAI-ECAI 2026 workshop), Ryan Marr (master's, co-author of the same paper), Runhui Xu (master's, second author, ACL 2026)
Reviewing	ICLR 2024–2026, AAAI 2024–2026, EMNLP 2025, NAACL 2027, AAMAS 2025–2027, ICML 2026, NeurIPS 2026
Teaching	Course Assistant at Columbia for Applied Computer Vision (Spring 2023) and Introduction to NLP (Summer and Fall 2022)

Open Source & Datasets

Data & Code	GameplayQA and X-Ego-CS on Hugging Face. Code for GameplayQA, PersonaEvolve, X-Ego, and DECOY on GitHub (HATS-ICT)
Tools	sync-video-label, a synchronized multi-POV video annotation tool with VLM pre-labeling, used to build GameplayQA
Packages	persona-factory (PyPI), a generator of coherent synthetic persona pools for LLM agents and social simulation
Community	ArcForge, an open-source game knowledge-graph explorer (110+ stars, 20+ contributors)

Skills

Methods	Diffusion transformers, flow matching, diffusion forcing, causal 3D video VAEs, contrastive learning (SigLIP / InfoNCE, GradCache), PPO / GRPO / IPPO / MAPPO, GNNs, LLM-as-judge evaluation, prompt and persona optimization, agent memory
ML	PyTorch, PyTorch Lightning, JAX, Hugging Face Transformers / Diffusers, PyG, scikit-learn, Weights & Biases, with V-JEPA2, DINOv2/v3, CLIP / SigLIP2, and Whisper encoders
LLM / Agents	OpenAI, Anthropic, and Gemini APIs, OpenRouter, LiteLLM, DSPy, LangChain, MCP, and the Claude Code and Codex harnesses
Simulation	Unity (C#, NavMesh), Unreal Engine 5 (C++, MetaHuman, UnrealCV), three.js / React Three Fiber, Panda3D + Bullet, Gymnasium, PettingZoo, ViZDoom, PyBullet, Blender
Engineering	Python, TypeScript / JavaScript, C++, C#, SQL, Bash, SLURM / HPC (A100, A40), Docker, AWS (Bedrock, EC2, S3), FastAPI, Flask, Next.js / React, Node.js, pandas / Polars, FFmpeg / OpenCV, pytest, CI
Languages	English (fluent), Chinese (native)